

# Supplementary ~~Material~~material: A formal knowledge base for ~~metagenomics~~amplicon sequencing data and geochemistry of the Rub' al Khali desert

## S1 Knowledge-representation patterns

This section ~~specifies the knowledge-representation patterns that are not reproduced~~gives formal patterns and connected graph examples for the resource described in the main article: ~~sampling, protocol execution, taxonomic abundance, XRF geochemistry, the sequencing chain and sequencing quality control, and adds a worked example of the main article's laboratory-control pattern.~~ Each pattern is given as its Description Logic axioms followed by a worked Turtle example taken from the released modules. Every printed triple and every prefixed IRI in this section is resolved against the staged modules by ~~scripts/validation/verify-manuscript-listings.py~~, which fails if a ~~listing drifts from the graph.~~ The Description Logic notation [1] uses  $\sqsubseteq$  for subclass inclusion,  $\sqcap$  for conjunction,  $\exists R.C$  for at least one relation  $R$  to a member of class  $C$ , and  $\forall R.C$  for restriction of all  $R$  fillers to  $C$ . The displayed patterns describe the intended entity relationships; structural admission is checked separately.

~~Listings use the prefix declarations of~~ Prepend Table S1; ~~prepend them to any listing to be executed against a local or deployed graph.~~ The `rak_kb:` prefix ~~covers the 's declarations when parsing the Turtle snippets.~~ The unprefixes knowledge-base namespace ~~without the RAK\_ local-name prefix, which is needed only for the PowerSoil extraction protocol~~`rak_kb:DNA_ext_powersoil`, the one individual in the released graph ~~minted without it~~ is used by the PowerSoil protocol. The listing validator checks examples against the available module snapshot; full validation additionally requires the corresponding taxonomy payload and pinned reference terms.

### S1.1 Laboratory-control worked example

~~Listing 1 instantiates the product-specification~~ The control module represents the whole-cell D6300 product, its composition specification and evidence-bearing expected-taxon pattern for the whole-cell D6300 standard. The promoted specification-to-taxon statement supports direct queries, while the reified assertion preserves its assertions as separate entities. The specification links directly to each supported taxon; a reified assertion additionally preserves the subject,

**Table S1** Prefix declarations used in all Turtle and SPARQL listings.

| Prefix     | Namespace                                   |
|------------|---------------------------------------------|
| rak:       | https://rubalkhali.science/kb/RAK_          |
| rak_kb:    | https://rubalkhali.science/kb/              |
| sio:       | http://semanticscience.org/resource/SIO_    |
| owl:       | http://www.w3.org/2002/07/owl#              |
| pato:      | http://purl.obolibrary.org/obo/PATO_        |
| uo:        | http://purl.obolibrary.org/obo/UO_          |
| envo:      | http://purl.obolibrary.org/obo/ENVO_        |
| ncbitaxon: | http://purl.obolibrary.org/obo/NCBITaxon_   |
| obo:       | http://purl.obolibrary.org/obo/             |
| rdf:       | http://www.w3.org/1999/02/22-rdf-syntax-ns# |
| rdfs:      | http://www.w3.org/2000/01/rdf-schema#       |
| xsd:       | http://www.w3.org/2001/XMLSchema#           |
| geo:       | http://www.opengis.net/ont/geosparql#       |
| dcterms:   | http://purl.org/dc/terms/                   |

relation~~descriptor~~, taxon, status and ~~evidencesource~~. Manufacturer genomic-DNA ~~percentages remain mass fractions are retained~~ in metadata/controls/control\_ composition.tsv; ~~the RDF does not reinterpret them as expected 16S read fractions.~~ Exact occurrence identifiers and their source links are distributed in ontology/ rubalkhali\_controls.ttl. This keeps long machine identifiers in the executable resource while exposing the scientific relationships here.

## S1.2 Sampling pattern

A sampling event is a process that derives a sample from a feature of interest (the site).

SamplingProcess  $\sqsubseteq$  Process (S1)

Sample  $\sqsubseteq$  MaterialEntity (S2)

SampleCollection  $\sqsubseteq$  Collection  $\sqcap \forall$ hasMember.Sample (S3)

SamplingProcess  $\sqsubseteq \exists$ hasTarget.Site  $\sqcap \exists$ hasOutput.SampleCollection (S4)

Sample  $\sqsubseteq \exists$ isDerivedFrom.Site (S5)

Listing S1 is generated from the Trip 1, Site 1 deep-soil record in the development graph.

**Listing S1** Turtle representation of a sampling process.

```
rak:P100001 a rak:0000024 ; # Sampling Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:7000001 . # hasOutput (Collection)

rak:7000001 a sio:001419 ; # Material sample collection
  sio:000059 rak:6000001 . # hasMember
```

```

rak:6000001 a rak:0000021 ; # Deep Soil Sample
sio:000244 rak:1000001 . # isDerivedFrom

```

~~Read in prose, an~~ The instance `rak:P100001` of a sampling process targets the site `rak:1000001` and yields a collection `rak:7000001` as its output. The collection explicitly contains `rak:6000001`, and this deep-soil sample is asserted to be derived from the same site. The redundancy between `hasTarget` on the process and `isDerivedFrom` on the sample is intentional: the former records the action, the latter records the resulting provenance edge directly on the material entity, so queries can traverse from sample to site without joining through the sampling event.

### S1.3 Experimental protocol pattern

~~We model experimental protocols as processes specified by a protocol entity~~ A protocol is a reusable information specification. A protocol execution is a process specified by that protocol. Each process (e.g., DNA extraction, library preparation) consumes inputs and produces outputs, explicitly linking agents and devices. The pattern instantiates SIO’s general *Process-hasInput/hasOutput-Object* idiom and adds a specification link to a reusable protocol description.

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \quad (\text{S6})$$

$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \quad (\text{S7})$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{isSpecifiedBy}.\text{Protocol} \quad (\text{S8})$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{hasInput}.\text{Entity} \sqcap \exists \text{hasOutput}.\text{Entity} \quad (\text{S9})$$

Listing S2 demonstrates the formalization of a DNA extraction process.

**Listing S2** Turtle representation of a DNA extraction protocol.

```

rak:P200001 a sio:000994 ; # Scientific Protocol Execution
sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
sio:000230 rak:6000001 ; # hasInput (Soil Sample)
sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:6800001 a rak:0000040 ; # DNA Extract
sio:000008 rak:5900001 . # hasAttribute (Concentration Quality)

```

~~In words, the~~ The protocol-execution instance `rak:P200001` is specified by the named extraction protocol `rak_kb:DNA_ext_powersoil`, takes the soil sample `rak:6000001` as input, and produces the DNA extract `rak:6800001` as output; the resulting extract carries a concentration quality (`rak:5900001`, class `rak:0000043`) which is itself quantified through a separate measurement process following the ~~pattern of the measurement pattern of~~ measurement pattern in the main article’s Knowledge representation subsection. The measurement value that quantifies it is the separate individual `rak:4900001` (class `rak:0000044`), which points to the quality with `sio:000215` and back from it with `sio:000216`. ~~Value and quality are deliberately distinct individuals in every module; conflating them is the single most common way to~~

~~break the traversal described in the Data Records section of the main article~~[Separate value and quality individuals preserve the reciprocal measurement links.](#)

## S1.4 Taxonomic abundance pattern

~~Taxonomic profiles represent collections of taxon observations containing associated relative and absolute abundances. Instead of modeling the feature table as a flat matrix, we formalize the entire table as a bioinformatic workflow that produces a dataset containing individual taxonomic observations. Here, we use the conservative mapping (Methods, taxonomy normalization) to assign either a validated NCBI class, an unambiguous retained project class, or a deterministic lineage-scoped contextual class as the target of each observation. Unreviewed GTDB candidates are not asserted~~[contain read counts and relative read abundances aggregated by rank. The validated mapping assigns NCBI or project identifiers to each rank-truncated classifier lineage. Listing S3 shows how an aggregated taxon-rank observation from one profile is represented in the knowledge graph](#)[gives one such observation.](#)

For instance, the sequence analysis for a specific profile yields an absolute sequence count associated with an audited taxonomic class. We construct an abundance quality that is an attribute of both that taxon and the source sequence dataset, binding the count back to the biological workflow. Formally, the pattern instantiates SIO's *Collection-hasMember-Thing* and reciprocal *MeasurementValue-isMeasurementValueOf-Quality* idioms:

$$\text{BioinformaticWorkflow} \sqsubseteq \text{Process} \quad (\text{S10})$$

$$\text{TaxonomicDataset} \sqsubseteq \text{Collection} \quad (\text{S11})$$

$$\text{BioinformaticWorkflow} \sqsubseteq \exists \text{hasInput.SequenceDataset} \sqcap \exists \text{hasOutput.TaxonomicDataset} \quad (\text{S12})$$

$$\text{TaxonomicDataset} \sqsubseteq \forall \text{hasMember.AbundanceObservation} \quad (\text{S13})$$

$$\text{AbundanceObservation} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AbundanceQuality} \quad (\text{S14})$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{isAttributeOf.Taxon} \quad (\text{S15})$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{hasMeasurementValue.AbundanceObservation} \quad (\text{S16})$$

**Listing S3** Turtle representation of taxonomic abundance formalized from a feature table.

```
rak:P290001 a rak:0000071 ; # Bioinformatic Workflow
  sio:000339 rak:L000012 ; # 16S amplicon processing protocol
  sio:000230 rak:7791009 ; # hasInput (FASTQ Dataset, ERR16783308)
  sio:000229 rak:7740001 . # hasOutput (Taxonomic Dataset)

rak:7740001 a rak:0000074 ; # taxon absolute abundance dataset
  sio:000059 rak:V00000001 . # hasMember (Observation Value)

rak:V00000001 a rak:0000076 ; # Absolute abundance value
  sio:000215 rak:Q00000001 ; # isMeasurementValueOf
  rak:2000025 "Domain: Bacteria" ; # classifier lineage
  rak:2000026 "853814.0"^^xsd:double . # absolute read count

rak:Q00000001 a rak:0000078, pato:0000070 ; # Absolute Abundance Quality
  sio:000011 ncbitaxon:2, rak:7791009 ; # taxon and FASTQ bearers
  sio:000216 rak:V00000001 . # hasMeasurementValue
```

~~Verbalized, the~~ The bioinformatic-workflow instance `rak:P290001` consumes the FASTQ dataset of run ERR16783308 (`rak:7791009`, profile V10Dr1) and produces the absolute taxonomic dataset `rak:7740001`; that dataset has as one of its members the measurement value `rak:V00000001`, which records an absolute read count of 853,814 for the domain-rank record `Domain: Bacteria`. Its reciprocal quality `rak:Q00000001` is an attribute of both *Bacteria* (`ncbitaxon:2`) and the source FASTQ dataset. Each of the fourteen datasets produced by one workflow instance holds a single rank at one of the two abundance bases, so a rank-restricted query never has to filter observations by lineage depth. ~~One aggregated~~ Each ~~taxon-rank entry is therefore expanded into a four-node sub-graph that preserves both the classifier lineage and its sequence-dataset provenance~~ adds a value and a quality while referencing shared workflow, dataset and taxon objects. Contextual lineages use the same pattern with their audited project IRI.

## S1.5 XRF geochemistry pattern

We model every XRF session as a process with multiple concentration-value outputs and a workflow-specific protocol. A laboratory session has an archived soil specimen as input. The field instrument log ~~, by contrast,~~ identifies a test and site ~~but not a physical specimen; the regenerated graph therefore links the session to the recorded sampling site and does not fabricate a collected specimen input, and the graph links each session to that site~~. Field-XRF and laboratory-XRF records remain distinguishable even when they report the same analyte. Analyte cross-references to ChEBI [2] and PubChem [3] are maintained in a single mapping ledger. The ~~fail-closed identifier~~ audit resolves every non-null identifier against pinned ChEBI v247 and a dated PubChem snapshot, checking identifier existence, label, formula, charge and atom-versus-oxide entity type. Of 93 instrument channels, 79 have a supported ChEBI cross-reference and 91 have a formula-matching PubChem cross-reference. ~~A missing external identifier remains explicitly null rather than being filled by label similarity. These are~~ Missing external identifiers remain explicitly null. The `rdfs:seeAlso` annotations ~~for the reporting channel, not assertions that a particular chemical species or mineral occurs in a sample. In particular, the instrument's~~ identify chemical references for reporting channels. The `P2O5` channel ~~is an~~ reports oxide-equivalent ~~reporting basis~~ magnitudes; its ChEBI cross-reference ~~has the same empirical ratio (P<sub>4</sub>O<sub>10</sub>) and is not a molecular-presence claim. The instrument's Light Elements (LE) output is an instrument pseudo-analyte or mixture, not an individual chemical entity, and is therefore not mapped to ChEBI or PubChem as a compound~~ shares the empirical ratio. Light Elements is an instrument-defined aggregate channel with null chemical cross-references.

The shared output pattern specializes the measurement pattern (~~the measurement pattern of the main article~~ main article, Knowledge representation); only the laboratory specialization requires a soil-sample input:

$$\text{XRFPProcess} \sqsubseteq \text{MeasurementProcess} \sqcap \exists \text{hasOutput.Concentration} \quad (\text{S17})$$

$$\text{LabXRFPProcess} \sqsubseteq \text{XRFPProcess} \sqcap \exists \text{hasInput.SoilSample} \quad (\text{S18})$$

$$\text{FieldXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasTarget.SamplingSite} \quad (\text{S19})$$

$$\text{Concentration} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AnalyteQuality} \quad (\text{S20})$$

Listing S4 is generated from the aluminium output of laboratory-XRF process rak:P300072 in the current module.

**Listing S4** Turtle representation of an XRF geochemistry measurement.

```
rak:P300072 a rak:0000025 ; # XRF Measuring Process
sio:000230 rak:6001732 ; # hasInput (sample V1Sr1)
sio:000339 rak:L_Surface_Lab ; # laboratory protocol
sio:000229 rak:4302121 . # hasOutput

rak:4302121 a rak:0000507 ; # Al concentration value
sio:000215 rak:5302121 ; # isMeasurementValueOf
rak:2000012 "0.78"^^xsd:double . # Concentration

rak:5302121 a rak:0000107 ; # Al concentration quality
sio:000011 rak:6001732 ; # isAttributeOf
sio:000216 rak:4302121 . # hasMeasurementValue
```

~~In words, this~~ This laboratory-XRF process consumes archived specimen rak:6001732 (sample V1Sr1) and produces an aluminium concentration value. The value points to its aluminium quality using sio:000215; the quality points back using sio:000216. A dedicated scan of every XRF source artifact (the processed laboratory table, the field log and all raw instrument exports) (`scripts/xrf/audit_xrf_unit_evidence.py`) finds no unit declaration attached to any concentration column; the only unit token in the exports is PPM, and it belongs to the lower-limit-of-detection column. The generated module therefore carries no unit assertion for XRF concentrations. ~~Readers should treat them as unitless instrument readings on a consistent scale within a workflow and must not use them for absolute cross-workflow comparisons. In the regenerated module, field~~ These values are instrument-reported magnitudes with undocumented concentration units. Field sessions use sio:000291 ~~(has target)~~ to identify the recorded to identify their site and retain the instrument test identifier; ~~they have no constructed “field-deep-soil-material” laboratory sessions identify their~~ specimen input. This distinction also prevents a shared graph pattern from implying numerical interchangeability between workflows Cross-workflow calibration requires documented units, paired aliquots and reference materials.

## S1.6 Sequencing workflow pattern

Sequencing data are produced by a chain of laboratory and computational processes that the knowledge graph represents as a sequence of *Process-hasInput/hasOutput-Object* steps: a DNA extraction process consumes a soil sample and produces a DNA extract; a library preparation process consumes the DNA extract and produces an amplicon library; a sequencing process consumes the amplicon library and produces a FASTQ dataset whose members are sequence reads. Each process is specified by a protocol and may carry agents and device participants, reusing the experimental protocol pattern ~~of the protocol pattern above~~ in Equation S6.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \quad (\text{S21})$$

$$\begin{aligned}
\text{DNAExtractionProcess} &\sqsubseteq \exists \text{hasInput}.\text{SoilSample} \sqcap \exists \text{hasOutput}.\text{DNAExtract} & (S22) \\
\text{LibraryPreparationProcess} &\sqsubseteq \text{ProtocolExecution} & (S23) \\
\text{LibraryPreparationProcess} &\sqsubseteq \exists \text{hasInput}.\text{DNAExtract} \sqcap \exists \text{hasOutput}.\text{AmpliconLibrary} & (S24) \\
\text{SequencingProcess} &\sqsubseteq \text{ProtocolExecution} & (S25) \\
\text{SequencingProcess} &\sqsubseteq \exists \text{hasInput}.\text{AmpliconLibrary} \sqcap \exists \text{hasOutput}.\text{FASTQDataset} & (S26) \\
\text{FASTQDataset} &\sqsubseteq \text{Dataset} & (S27) \\
\text{FASTQDataset} &\sqsubseteq \forall \text{hasMember}.\text{SequenceRead} & (S28)
\end{aligned}$$

Listing S5 instantiates the complete sequencing chain.

**Listing S5** Turtle representation of the sequencing workflow from soil sample T1Dr1 to ENA run ERR16061083.

```

rak:P200923 a sio:000994 ; # DNA Extraction Process
  sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000577 ; # hasInput (T1Dr1 Soil Sample)
  sio:000229 rak:6800540 . # hasOutput (DNA Extract)

rak:P250001 a rak:0000065 ; # Library Preparation Process
  sio:000339 rak:L000010 ; # Illumina 16S Prep Guide
  sio:000230 rak:6800540 ; # hasInput (DNA Extract)
  sio:000229 rak:6690001 . # hasOutput (Amplicon Library)

rak:P280001 a rak:0000066 ; # Sequencing Process
  sio:000230 rak:6690001 ; # hasInput (Amplicon Library)
  sio:000229 rak:7790001 . # hasOutput (FASTQ Dataset)

rak:7790001 a rak:0000063 ; # FASTQ Dataset
  rdfs:seeAlso <https://identifiers.org/insdc.run/ERR16061083> .

```

~~Read as a chain, the~~ The DNA extraction process `rak:P200923` consumes the soil sample `rak:6000577` and produces the DNA extract `rak:6800540`; that extract is the input of the library preparation `rak:P250001`, which yields the amplicon library `rak:6690001`; finally the sequencing process `rak:P280001` consumes the library and produces the FASTQ dataset `rak:7790001`, which is cross-referenced to its ENA accession via `rdfs:seeAlso`. Each link in the chain is the same `hasInput/hasOutput` pattern, so a single SPARQL traversal can recover the recorded provenance from any node in this chain.

## S1.7 Sequencing quality control pattern

Sequencing quality control is represented as an additional computational process that takes a FASTQ dataset as input and produces multiple measurement values, each instantiating the ~~generic measurement pattern of the measurement pattern of measurement pattern in~~ the main article's Knowledge representation subsection. Metrics (sequence count, GC content, duplicate rate, read length) are split into forward (R1) and reverse (R2) subclasses so that the provenance of paired-end runs is preserved.

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \quad (S29)$$

SequencingQCProcess  $\sqsubseteq \exists \text{hasInput.FASTQDataset} \sqcap \exists \text{hasOutput.QCMeasurementValue}$  (S30)

QCMeasurementValue  $\sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.QCQuality}$  (S31)

QCQuality  $\sqsubseteq \exists \text{isAttributeOf.FASTQDataset}$  (S32)

Listing S6 shows the corresponding quality-control pattern.

**Listing S6** Turtle representation of forward-read GC content for ENA run ERR16062319.

```
rak:P350001 a rak:0000200 ; # Sequencing QC Process
sio:000230 rak:7790263 ; # hasInput (FASTQ Dataset)
sio:000229 rak:4450001 . # hasOutput (QC Value)

rak:4450001 a rak:0000217 ; # Forward GC Content Value
sio:000215 rak:5550001 ; # isMeasurementValueOf
sio:000221 uo:0000187 ; # Percent
rak:2000074 "58.0"^^xsd:double . # has GC content

rak:5550001 a rak:0000205 ; # Forward GC Content Quality
sio:000011 rak:7790263 ; # isAttributeOf (FASTQ Dataset)
sio:000216 rak:4450001 . # hasMeasurementValue
```

In words, the QC process `rak:P350001` takes the FASTQ dataset `rak:7790263` as input and produces the measurement value `rak:4450001`, which records a GC content of 58.0 percent and is the measurement value of a forward-read GC-content quality `rak:5550001` that is in turn an attribute of the same FASTQ dataset. Splitting the quality class into forward and reverse subclasses preserves the read direction, so that paired-end QC metrics can be queried independently per orientation.

## S2 ~~Supplementary Tables~~ Library preparation and archived-soil pH

First-round PCR used 50  $\mu\text{L}$  reactions with  $1 \times \text{Mg}^{2+}$ -containing buffer, 0.2 mM of each dNTP, 0.4  $\mu\text{M}$  of each primer and 1.25 U of TaKaRa Ex Taq. The programme comprised  $95^\circ\text{C}$  for 3 min, followed by 35 cycles of  $95^\circ\text{C}$  for 10 s,  $65^\circ\text{C}$  for 30 s and  $72^\circ\text{C}$  for 1 min, with a final extension at  $75^\circ\text{C}$  for 5 min. Initial DNA input was 5  $\mu\text{L}$ ; repeat attempts used 8 and 12  $\mu\text{L}$  while retaining the 50  $\mu\text{L}$  total reaction volume. Indexed libraries underwent 0.8:1 AMPure XP cleanup. The retained indexing protocol specifies the kit and thermal programme; its reaction-specific reagent volumes remain undocumented.

Pooling used library molarity derived from measured DNA concentration and fragment length. The retained template calculates aliquot volumes using a pooling-volume factor of the form  $VC/N$ , where  $V$  is target pool volume,  $C$  is target molarity and  $N$  is the number of libraries; dividing this factor by a library's molarity gives that library's aliquot volume. The retained template leaves the target values and library measurements blank. The actual pooling volumes therefore require the completed run-specific sheet.

Both PMA-treated and untreated aliquots underwent 10 min in darkness and 35 min of illumination at 450 nm. Each of three parent collection tubes supplied three

separately prepared slurries, with each slurry split into a treated and an untreated aliquot.

Archived-soil pH suspensions were mixed for 30 min on a HulaMixer, then stood for 30 min at room temperature. Measurements used an accumet AB315 meter and Fisher Scientific accumet electrode 13-620-112, with the green stability indicator as the reading endpoint. All Trip 4 assays used archived field replicate 2. The EQ-PH-SHARED-v1.0.1 derivative links admitted measurements to the confirmed specimens and preserves the original workbook identifiers and assay values.

### S3 XRF acquisition evidence

The source workbooks retain 547 selected specimens from Trips 1–4 and 178 from Trip 5. All laboratory measurements used Fast Mode, as confirmed by the project lead on 9 September 2026. For Trip 5, each selected specimen has the workbook settings Fast Screening-He8mm, He, 8mm and Oxides. The deep-soil workbook contains 58 ordinary specimen sheets and two method comparisons, for V4Dr3 and V18Dr3. The parser retains the Fast Screening sheet from each comparison, giving 60 deep-soil specimens. The surface and root-adjacent workbooks contribute 60 and 58 specimens, respectively. The two Best Detection sheets remain in the original workbook.

The Trips 1–4 consolidation preserves 28 export columns. Its index names the pre-consolidation workbooks, while acquisition metadata rows remain unavailable. The all-campaign confirmation establishes the Fast Mode method family; the exact method label, gas and diameter for Trips 1–4 remain undocumented. The handheld field exports specify Geochem(3-Beam) and NORMAL, with duration recorded as N/A. These field settings belong to the Olympus Vanta analyser.

Concentration and calculated-concentration columns retain their original numeric magnitudes. Trip 5 uses the spreadsheet format General for these columns. Percent formats occur in the statistical-error columns and four lower-limit-of-detection cells; PPM labels occur in lower-limit-of-detection columns. Concentration units, laboratory instrument identity, exposure settings, sample preparation and calibration remain undocumented in the recovered workbooks. The sheet-level audit, source checksums and project-lead confirmation are linked from docs/XRF\_ACQUISITION.md. The command `scripts/xrf/audit_xrf_workbook_metadata.py` reproduces the workbook audit; its regression tests check all 178 selected Trip 5 specimens, method-comparison selection and unit evidence.

### S4 Supplementary tables

Table S2 reports the curated field observations, and Table S3 reports the audited XRF channel cross-references.

**Table S2:** Site-specific field environmental observations across all campaigns. Values are generated from the immutable source sheets and the versioned correction ledger; a dash denotes an unrecorded or quarantined value. Altitude was not recorded in these field sheets.

| Expedition    | Site | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------|------------|-------|----------------------|------------|---------------|--------|
| Trip 1 (2023) | 1    | 17/03/2023 | 07:58 | 19.1498 N, 45.1665 E | 20.7       | 928           | 62.5   |
| Trip 1 (2023) | 2    | 17/03/2023 | 08:21 | 19.2035 N, 45.2448 E | 21.3       | 928           | 60.4   |
| Trip 1 (2023) | 3    | 17/03/2023 | 08:50 | 19.3011 N, 45.3844 E | 22.6       | 928           | 55.4   |
| Trip 1 (2023) | 4    | 17/03/2023 | 09:20 | 19.3796 N, 45.5518 E | 24         | 928           | 51.0   |
| Trip 1 (2023) | 5    | 17/03/2023 | 10:29 | 19.4072 N, 45.6595 E | 28.8       | 928           | 40.6   |
| Trip 1 (2023) | 6    | 17/03/2023 | 11:07 | 19.4057 N, 45.8161 E | 29         | 928           | 42.9   |
| Trip 1 (2023) | 7    | 17/03/2023 | 11:35 | 19.4070 N, 45.9539 E | 28.7       | 928           | 49.4   |
| Trip 1 (2023) | 8    | 17/03/2023 | 12:05 | 19.3932 N, 46.1107 E | 30.6       | 928           | 41.5   |
| Trip 1 (2023) | 9    | 17/03/2023 | 12:44 | 19.4112 N, 46.3027 E | 35.9       | 928           | 36.3   |
| Trip 1 (2023) | 10   | 17/03/2023 | 13:12 | 19.4205 N, 46.4608 E | 32.5       | 928           | 39.0   |
| Trip 1 (2023) | 11   | 17/03/2023 | 13:39 | 19.4444 N, 46.6356 E | 31.1       | 930           | 37.5   |
| Trip 1 (2023) | 12   | 17/03/2023 | 14:04 | 19.4779 N, 46.8344 E | 34.6       | 932           | 30.0   |
| Trip 1 (2023) | 13   | 17/03/2023 | 15:42 | 19.5293 N, 47.0533 E | 28.9       | 936           | 29.2   |
| Trip 1 (2023) | 14   | 17/03/2023 | 16:09 | 19.5813 N, 47.2920 E | 30.2       | 938           | 37.8   |
| Trip 1 (2023) | 15   | 17/03/2023 | 16:33 | 19.6137 N, 47.4759 E | 31.6       | 946           | 33.0   |
| Trip 1 (2023) | 16   | 17/03/2023 | 16:55 | 19.6445 N, 47.6261 E | 20.7       | 948           | 33.3   |
| Trip 1 (2023) | 17   | 17/03/2023 | 17:23 | 19.6765 N, 47.8273 E | 20.5       | 952           | 32.7   |
| Trip 1 (2023) | 18   | 18/03/2023 | 06:49 | 19.7325 N, 48.0469 E | 20         | 954           | 50.0   |
| Trip 1 (2023) | 19   | 18/03/2023 | 07:15 | 19.8123 N, 48.2405 E | 22.9       | 960           | 51.4   |
| Trip 1 (2023) | 20   | 18/03/2023 | 07:55 | 19.8693 N, 48.4246 E | 24.3       | 964           | 39.6   |
| Trip 1 (2023) | 21   | 18/03/2023 | 08:20 | 19.9011 N, 48.4865 E | 29.4       | 962           | 30.9   |
| Trip 1 (2023) | 22   | 18/03/2023 | 08:45 | 19.9787 N, 48.6394 E | 29         | —             | 25.9   |
| Trip 1 (2023) | 23   | 18/03/2023 | 09:25 | 20.0493 N, 48.7733 E | 26.8       | 966           | 25.2   |
| Trip 1 (2023) | 24   | 18/03/2023 | 09:48 | 20.0927 N, 48.8994 E | 29.5       | 966           | 25.9   |
| Trip 1 (2023) | 25   | 18/03/2023 | 10:15 | 20.1003 N, 48.9542 E | 27.8       | 968           | 27.6   |
| Trip 1 (2023) | 26   | 18/03/2023 | 10:40 | 20.1488 N, 49.0787 E | 31.7       | 968           | 23.1   |
| Trip 1 (2023) | 27   | 18/03/2023 | 11:18 | 20.2039 N, 49.1960 E | 31.5       | 970           | 23.8   |
| Trip 1 (2023) | 28   | 18/03/2023 | 11:47 | 20.2438 N, 49.3709 E | 31.9       | 970           | 24.9   |
| Trip 1 (2023) | 29   | 18/03/2023 | 12:15 | 20.2598 N, 49.5675 E | 31.7       | 968           | 25.7   |
| Trip 1 (2023) | 30   | 18/03/2023 | 14:39 | 20.2693 N, 49.6837 E | 34.1       | 964           | 22.4   |
| Trip 1 (2023) | 31   | 18/03/2023 | 15:11 | 20.2946 N, 49.8914 E | 33.8       | 964           | 22.3   |

Continued on next page

| Expedition    | Site       | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------------|------------|-------|----------------------|------------|---------------|--------|
| Trip 1 (2023) | 32         | 18/03/2023 | 15:44 | 20.3107 N, 50.0930 E | 34         | 968           | 22.0   |
| Trip 1 (2023) | 33         | 18/03/2023 | 16:15 | 20.3175 N, 50.3031 E | 34.8       | 968           | 20.1   |
| Trip 1 (2023) | 34         | 18/03/2023 | 16:46 | 20.3444 N, 50.5154 E | 34.1       | 970           | 21.1   |
| Trip 1 (2023) | 35         | 18/03/2023 | 17:24 | 20.3718 N, 50.7300 E | 33.7       | 976           | 20.8   |
| Trip 1 (2023) | 36         | 19/03/2023 | 09:40 | 20.4232 N, 51.1448 E | 29.8       | 982           | 15.9   |
| Trip 1 (2023) | 37         | 19/03/2023 | 10:10 | 20.4542 N, 51.3449 E | 33.5       | 984           | 14.4   |
| Trip 1 (2023) | 38         | 19/03/2023 | 10:30 | 20.4610 N, 51.3702 E | 31.9       | 984           | 15.1   |
| Trip 1 (2023) | 39         | 19/03/2023 | 11:05 | 20.5102 N, 51.5974 E | 36.4       | 984           | 16.0   |
| Trip 1 (2023) | 40         | 19/03/2023 | 11:40 | 20.5472 N, 51.8256 E | 30.5       | 980           | 30.4   |
| Trip 1 (2023) | 41         | 19/03/2023 | 12:16 | 20.5398 N, 51.9461 E | 35.4       | 980           | 19.8   |
| Trip 1 (2023) | 42         | 19/03/2023 | 12:41 | 20.5552 N, 52.1240 E | 35.4       | 984           | 20.5   |
| Trip 1 (2023) | 43         | 19/03/2023 | 13:05 | 20.5676 N, 52.2241 E | 35.2       | 984           | 22.8   |
| Trip 1 (2023) | 44         | 19/03/2023 | 13:40 | 20.5678 N, 52.4104 E | 38.4       | 984           | 19.6   |
| Trip 1 (2023) | 45         | 19/03/2023 | 13:59 | 20.5635 N, 52.4678 E | 37.1       | 986           | 20.1   |
| Trip 1 (2023) | 46         | 19/03/2023 | 14:29 | 20.5537 N, 52.6692 E | 35.7       | 987           | 21.1   |
| Trip 1 (2023) | 47         | 19/03/2023 | 15:00 | 20.5418 N, 52.8688 E | 37.3       | 986           | 20.3   |
| Trip 1 (2023) | 48         | 19/03/2023 | 17:00 | 20.5686 N, 53.0940 E | 35.3       | 984           | 23.7   |
| Trip 1 (2023) | 49         | 19/03/2023 | 17:30 | 20.6447 N, 53.2879 E | 35         | 986           | 21.7   |
| Trip 1 (2023) | 50         | 20/03/2023 | 07:14 | 20.7139 N, 53.4123 E | 29.8       | 988           | 31.3   |
| Trip 1 (2023) | 51         | 20/03/2023 | 07:37 | 20.7987 N, 53.4870 E | 30.2       | 992           | 32.5   |
| Trip 1 (2023) | 52         | 20/03/2023 | 08:02 | 20.8278 N, 53.5784 E | 31.1       | 992           | 29.7   |
| Trip 1 (2023) | 53         | 20/03/2023 | 09:03 | 20.8515 N, 53.7579 E | 34.8       | 992           | 26.0   |
| Trip 1 (2023) | 54         | 20/03/2023 | 10:03 | 20.9202 N, 53.9622 E | 32.1       | 992           | 26.7   |
| Trip 1 (2023) | 55         | 20/03/2023 | 10:24 | 20.9023 N, 54.0024 E | 33.2       | 990           | 27.9   |
| Trip 1 (2023) | 56         | 20/03/2023 | 11:05 | 20.8216 N, 54.2239 E | 30         | 988           | 35.5   |
| Trip 1 (2023) | 57         | 20/03/2023 | 11:28 | 20.7350 N, 54.4176 E | 31.3       | 988           | 38.1   |
| Trip 1 (2023) | 58         | 20/03/2023 | 11:50 | 20.6654 N, 54.5933 E | 33.4       | 986           | 39.4   |
| Trip 1 (2023) | 59         | 20/03/2023 | 12:04 | 20.6563 N, 54.6359 E | 37.1       | 986           | 43.1   |
| Trip 1 (2023) | 60         | 20/03/2023 | 12:49 | 20.7064 N, 54.7321 E | 37.1       | 986           | 34.5   |
| Trip 1 (2023) | water well | 20/03/2023 | 13:16 | 20.7159 N, 54.7997 E | 35.6       | 984           | 31.4   |
| Trip 1 (2023) | road       | 20/03/2023 | 13:53 | 20.8258 N, 54.9889 E | 35.1       | 982           | 32.2   |
| Trip 1 (2023) | road       | 20/03/2023 | 14:20 | 20.9341 N, 55.1261 E | 36         | 982           | 26.3   |
| Trip 1 (2023) | camground  | 20/03/2023 | 14:45 | 21.0311 N, 55.1598 E | 37         | 984           | 26.0   |
| Trip 2 (2023) | 1          | 14/07/2023 | 06:10 | 19.1498 N, 45.1665 E | 34.5       | —             | —      |
| Trip 2 (2023) | 2          | 14/07/2023 | 07:12 | 19.2035 N, 45.2448 E | 36.7       | —             | —      |

Continued on next page

| Expedition    | Site | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------|------------|-------|----------------------|------------|---------------|--------|
| Trip 2 (2023) | 3    | 14/07/2023 | 07:16 | 19.3011 N, 45.3844 E | 37.0       | —             | —      |
| Trip 2 (2023) | 4    | 14/07/2023 | 07:36 | 19.3796 N, 45.5518 E | 38.6       | —             | —      |
| Trip 2 (2023) | 5    | 14/07/2023 | 07:43 | 19.4072 N, 45.6595 E | 39.2       | —             | —      |
| Trip 2 (2023) | 6    | 14/07/2023 | 08:58 | 19.4057 N, 45.8161 E | 41.2       | —             | —      |
| Trip 2 (2023) | 7    | 14/07/2023 | 09:20 | 19.4070 N, 45.9539 E | 41.3       | —             | —      |
| Trip 2 (2023) | 8    | 14/07/2023 | 10:20 | 19.3932 N, 46.1107 E | 41.9       | —             | —      |
| Trip 3 (2024) | 1    | 17/02/2024 | 06:54 | 19.1498 N, 45.1665 E | 21.9       | 898           | 55     |
| Trip 3 (2024) | 2    | 17/02/2024 | 07:21 | 19.2035 N, 45.2448 E | 26.6       | 904           | 40.9   |
| Trip 3 (2024) | 3    | 17/02/2024 | 08:05 | 19.3011 N, 45.3844 E | 28.3       | 914           | 38.5   |
| Trip 3 (2024) | 4    | 17/02/2024 | 08:40 | 19.3796 N, 45.5518 E | 30.2       | 922           | 37.5   |
| Trip 3 (2024) | 5    | 17/02/2024 | 09:51 | 19.4072 N, 45.6595 E | 35.2       | 926           | 19.5   |
| Trip 3 (2024) | 6    | 17/02/2024 | 10:45 | 19.4057 N, 45.8161 E | 37.1       | 942           | 17.6   |
| Trip 3 (2024) | 7    | 17/02/2024 | 11:28 | 19.4070 N, 45.9539 E | 37         | 926           | 19.6   |
| Trip 3 (2024) | 8    | 17/02/2024 | 11:50 | 19.3932 N, 46.1107 E | 39         | 922           | 19     |
| Trip 3 (2024) | 9    | 17/02/2024 | 12:31 | 19.4112 N, 46.3027 E | 36.4       | 924           | 21.5   |
| Trip 3 (2024) | 10   | 17/02/2024 | 13:22 | 19.4205 N, 46.4608 E | 37.1       | 926           | 18     |
| Trip 3 (2024) | 11   | 17/02/2024 | 13:46 | 19.4444 N, 46.6356 E | 41.2       | 930           | 21     |
| Trip 3 (2024) | 12   | 17/02/2024 | 14:04 | 19.4779 N, 46.8344 E | 39.9       | 932           | 15     |
| Trip 3 (2024) | 13   | 17/02/2024 | 15:20 | 19.5293 N, 47.0533 E | 40         | 940           | 13.5   |
| Trip 3 (2024) | 14   | 17/02/2024 | 15:50 | 19.5813 N, 47.2920 E | 41.2       | 940           | 14.9   |
| Trip 3 (2024) | 15   | 17/02/2024 | 16:13 | 19.6137 N, 47.4759 E | 38.8       | 944           | 13.5   |
| Trip 3 (2024) | 16   | 17/02/2024 | 16:34 | 19.6445 N, 47.6261 E | 39.4       | 948           | 11.9   |
| Trip 3 (2024) | 17   | 18/02/2024 | 08:42 | 19.6765 N, 47.8273 E | 23         | 956           | 27     |
| Trip 3 (2024) | 18   | 18/02/2024 | 09:54 | 19.7325 N, 48.0469 E | 29.2       | 960           | 18.7   |
| Trip 3 (2024) | 19   | 18/02/2024 | 10:25 | 19.8123 N, 48.2405 E | 27.4       | 962           | 21.2   |
| Trip 3 (2024) | 20   | 18/02/2024 | 11:05 | 19.8693 N, 48.4246 E | 32.9       | —             | 17.7   |
| Trip 3 (2024) | 21   | 18/02/2024 | 12:08 | 19.9011 N, 48.4865 E | 31.3       | 964           | 31.321 |
| Trip 3 (2024) | 22   | 18/02/2024 | 12:52 | 19.9787 N, 48.6394 E | 31.3       | 968           | 21.3   |
| Trip 3 (2024) | 23   | 18/02/2024 | 13:35 | 20.0493 N, 48.7733 E | 24.4       | 966           | 30.3   |
| Trip 3 (2024) | 24   | 18/02/2024 | 14:18 | 20.0927 N, 48.8994 E | 25.5       | 966           | 31.3   |
| Trip 3 (2024) | 25   | 18/02/2024 | 14:38 | 20.1003 N, 48.9542 E | 31         | 966           | 24.3   |
| Trip 3 (2024) | 26   | 18/02/2024 | 15:12 | 20.1488 N, 49.0787 E | 31.9       | 966           | 25.6   |
| Trip 3 (2024) | 27   | 18/02/2024 | 15:45 | 20.2039 N, 49.1960 E | 32.4       | 968           | 23.7   |
| Trip 3 (2024) | 28   | 18/02/2024 | 16:24 | 20.2438 N, 49.3709 E | 27.4       | 968           | 27.8   |
| Trip 3 (2024) | 29   | 18/02/2024 | 16:56 | 20.2598 N, 49.5675 E | 27.3       | 968           | 27     |

Continued on next page

| Expedition    | Site           | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|----------------|------------|-------|----------------------|------------|---------------|--------|
| Trip 3 (2024) | 30             | 19/02/2024 | 08:10 | 20.2693 N, 49.6837 E | 36.8       | 976           | 20.2   |
| Trip 3 (2024) | 31             | 19/02/2024 | 08:43 | 20.2946 N, 49.8914 E | 23.1       | 978           | 29.3   |
| Trip 3 (2024) | 32             | 19/02/2024 | 09:20 | 20.3107 N, 50.0930 E | 26.7       | 982           | 23     |
| Trip 3 (2024) | 33             | 19/02/2024 | 10:07 | 20.3175 N, 50.3031 E | 26.6       | 984           | 20.9   |
| Trip 3 (2024) | 34             | 19/02/2024 | 10:52 | 20.3444 N, 50.5154 E | 29.9       | 982           | 17.7   |
| Trip 3 (2024) | 35             | 19/02/2024 | 11:30 | 20.3718 N, 50.7300 E | 28.7       | 984           | 17.9   |
| Trip 3 (2024) | 36             | 19/02/2024 | 13:58 | 20.4232 N, 51.1448 E | 28.2       | 982           | 16.1   |
| Trip 3 (2024) | 37             | 19/02/2024 | 14:52 | 20.4542 N, 51.3449 E | 29.1       | 984           | 15.5   |
| Trip 3 (2024) | 38             | 19/02/2024 | 15:01 | 20.4610 N, 51.3702 E | 29.3       | 984           | 15.1   |
| Trip 3 (2024) | 39             | 19/02/2024 | 15:56 | 20.5102 N, 51.5974 E | 28.8       | 984           | 16     |
| Trip 3 (2024) | 40             | 19/02/2024 | 16:42 | 20.5472 N, 51.8256 E | 28.2       | 986           | 17.3   |
| Trip 3 (2024) | 41             | 19/02/2024 | 17:14 | 20.5398 N, 51.9461 E | 29.4       | 988           | 16.2   |
| Trip 3 (2024) | 42             | 20/02/2024 | 08:15 | 20.5552 N, 52.1240 E | 24.6       | 990           | 34.2   |
| Trip 3 (2024) | 43             | 20/02/2024 | 08:38 | 20.5676 N, 52.2241 E | 27         | 992           | 27.8   |
| Trip 3 (2024) | 44             | 20/02/2024 | 09:22 | 20.5678 N, 52.4104 E | 29.1       | 994           | 22     |
| Trip 3 (2024) | 45             | 20/02/2024 | 09:44 | 20.5635 N, 52.4678 E | 32.4       | 996           | 18.3   |
| Trip 3 (2024) | 46             | 20/02/2024 | 10:44 | 20.5537 N, 52.6692 E | 32.3       | 994           | 17.4   |
| Trip 3 (2024) | 47             | 20/02/2024 | 11:24 | 20.5418 N, 52.8688 E | 33.2       | 994           | 17.2   |
| Trip 3 (2024) | 48             | 20/02/2024 | 12:54 | 20.5686 N, 53.0940 E | 49.4       | 994           | 9.5    |
| Trip 3 (2024) | 49             | 20/02/2024 | 14:21 | 20.6447 N, 53.2879 E | 34         | 994           | 13.4   |
| Trip 3 (2024) | 50             | 20/02/2024 | 14:51 | 20.7139 N, 53.4123 E | 31.5       | 996           | 21.6   |
| Trip 3 (2024) | 51             | 20/02/2024 | 16:11 | 20.7987 N, 53.4870 E | 35.9       | 994           | 16.5   |
| Trip 3 (2024) | 52             | 20/02/2024 | 06:57 | 20.8278 N, 53.5784 E | 34.3       | 994           | 17.5   |
| Trip 3 (2024) | 53             | 21/02/2024 | 06:06 | 20.8515 N, 53.7579 E | 21.9       | 990           | 35     |
| Trip 3 (2024) | 54             | 21/02/2024 | 15:58 | 20.9202 N, 53.9622 E | 32.2       | 998           | 23.5   |
| Trip 3 (2024) | 55             | 21/02/2024 | 15:41 | 20.9023 N, 54.0024 E | 31.3       | 998           | 14.3   |
| Trip 3 (2024) | 56             | 21/02/2024 | 15:10 | 20.8216 N, 54.2239 E | 31.6       | 998           | 13.5   |
| Trip 3 (2024) | 57             | 21/02/2024 | 14:48 | 20.7350 N, 54.4176 E | 34.2       | 996           | 16.3   |
| Trip 3 (2024) | 58             | 21/02/2024 | 14:29 | 20.6654 N, 54.5933 E | 35.6       | 992           | 16.4   |
| Trip 3 (2024) | 59             | 21/02/2024 | 14:15 | 20.6563 N, 54.6359 E | 36.8       | 992           | 14.3   |
| Trip 3 (2024) | 60             | 21/02/2024 | 13:20 | 20.7064 N, 54.7321 E | 37.1       | 992           | 14.3   |
| Trip 3 (2024) | oill spill     | 17/02/2024 | 10:25 | 19.4054 N, 45.7271 E | 35.1       | —             | 20     |
| Trip 3 (2024) | water well     | 17/02/2024 | 14:23 | 19.5014 N, 46.9015 E | 39.2       | —             | 15.9   |
| Trip 3 (2024) | Camell         | 17/02/2024 | 16:49 | 19.6540 N, 47.6883 E | 37.6       | —             | 25     |
| Trip 3 (2024) | plant gradient | 18/02/2024 | 09:25 | 19.7092 N, 47.9593 E | 25         | 958           | 24     |

Continued on next page

| Expedition                      | Site                     | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------------------------|--------------------------|------------|-------|----------------------|------------|---------------|--------|
| Trip 3 (2024)                   | hot spring               | 19/02/2024 | 12:25 | 20.3972 N, 50.9655 E | 32.1       | 984           | 14.8   |
| Trip 1 auxiliary/revisit (2023) | 12.5+A                   | 17/03/2023 | 14:22 | 19.5015 N, 46.9014 E | 33         | 934           | 30     |
| Trip 1 auxiliary/revisit (2023) | 19.5                     | 18/03/2023 | 07:20 | 19.8193 N, 48.2402 E | 21.8       | 960           | 61.9   |
| Trip 1 auxiliary/revisit (2023) | 27+                      | 18/03/2023 | 11:21 | 20.2039 N, 49.1960 E | 31.5       | 970           | 23.8   |
| Trip 1 auxiliary/revisit (2023) | 31urine                  | 18/03/2023 | 15:14 | 20.2946 N, 49.8914 E | 33.8       | 964           | 22.3   |
| Trip 1 auxiliary/revisit (2023) | 32+                      | 18/03/2023 | 15:44 | 20.3107 N, 50.0930 E | 34         | 968           | 22     |
| Trip 1 auxiliary/revisit (2023) | 41+                      | 19/03/2023 | 12:16 | 20.5398 N, 51.9461 E | 35.4       | 980           | 19.8   |
| Trip 1 auxiliary/revisit (2023) | 47+                      | 19/03/2023 | 15:00 | 20.5418 N, 52.8688 E | 37.3       | 986           | 20.3   |
| Trip 1 auxiliary/revisit (2023) | 47+grass                 | 19/03/2023 | 15:00 | 20.5418 N, 52.8688 E | 37.3       | 986           | 20.3   |
| Trip 1 auxiliary/revisit (2023) | 48+                      | 19/03/2023 | 17:00 | 20.5686 N, 53.0940 E | 35.3       | 984           | 23.7   |
| Trip 1 auxiliary/revisit (2023) | 48+                      | 19/03/2023 | 17:00 | 20.5686 N, 53.0940 E | 35.3       | 984           | 23.7   |
| Trip 1 auxiliary/revisit (2023) | 52AD                     | 20/03/2023 | 08:02 | 20.8278 N, 53.5784 E | 31.1       | 992           | 29.7   |
| Trip 1 auxiliary/revisit (2023) | 54F                      | 20/03/2023 | 10:16 | 20.9202 N, 53.9622 E | 34.8       | 990           | 26.9   |
| Trip 1 auxiliary/revisit (2023) | 35 + WaterWellRestStop   | 18/03/2023 | 18:48 | 20.3972 N, 50.9657 E | —          | 976           | —      |
| Trip 1 auxiliary/revisit (2023) | 35 + hot spring filter1  | 19/03/2023 | 08:03 | 20.3972 N, 50.9657 E | 28.3       | 980           | 17.7   |
| Trip 1 auxiliary/revisit (2023) | 35 + hot spring filter 2 | 19/03/2023 | 08:11 | 20.3972 N, 50.9657 E | 28.3       | 980           | 17.7   |
| Trip 4 (2024)                   | 1                        | 29/08/2024 | 06:04 | 19.1498 N, 45.1665 E | 30.83      | —             | 19.4   |
| Trip 4 (2024)                   | 2                        | 29/08/2024 | 06:24 | 19.2035 N, 45.2448 E | 31.5       | —             | 18.4   |
| Trip 4 (2024)                   | 3                        | 29/08/2024 | 06:53 | 19.3011 N, 45.3844 E | 33.61      | —             | 16.5   |
| Trip 4 (2024)                   | 4                        | 29/08/2024 | 07:15 | 19.3796 N, 45.5518 E | 35.11      | —             | 15.5   |
| Trip 4 (2024)                   | 5                        | 29/08/2024 | 07:29 | 19.4072 N, 45.6595 E | 36.67      | —             | 14     |
| Trip 4 (2024)                   | 6                        | 29/08/2024 | 08:07 | 19.4057 N, 45.8161 E | 37.06      | —             | 13.3   |
| Trip 4 (2024)                   | 7                        | 29/08/2024 | 08:32 | 19.4070 N, 45.9539 E | 37.56      | —             | 13     |
| Trip 4 (2024)                   | 8                        | 29/08/2024 | 08:57 | 19.3932 N, 46.1107 E | 37.39      | —             | 14.2   |
| Trip 4 (2024)                   | 9                        | 29/08/2024 | 09:28 | 19.4112 N, 46.3027 E | 39         | —             | 13.5   |
| Trip 4 (2024)                   | 10                       | 29/08/2024 | 09:50 | 19.4205 N, 46.4608 E | 39.56      | —             | 13.4   |
| Trip 4 (2024)                   | 11                       | 29/08/2024 | 10:14 | 19.4444 N, 46.6356 E | 42         | —             | 12.4   |
| Trip 4 (2024)                   | 12                       | 29/08/2024 | 10:35 | 19.4779 N, 46.8344 E | 42.94      | —             | 21.5   |
| Trip 4 (2024)                   | 13                       | 29/08/2024 | 11:30 | 19.5293 N, 47.0533 E | 41.94      | —             | 16.5   |
| Trip 4 (2024)                   | 14                       | 29/08/2024 | 11:50 | 19.5813 N, 47.2920 E | 42.56      | —             | 18     |
| Trip 4 (2024)                   | 15                       | 29/08/2024 | 12:06 | 19.6137 N, 47.4759 E | 44.11      | —             | 15.2   |
| Trip 4 (2024)                   | 16                       | 29/08/2024 | 12:21 | 19.6445 N, 47.6261 E | 43.5       | —             | 15.7   |
| Trip 4 (2024)                   | 17                       | 29/08/2024 | 12:49 | 19.6765 N, 47.8273 E | 43.11      | —             | 15.2   |
| Trip 4 (2024)                   | 18                       | 29/08/2024 | 13:24 | 19.7325 N, 48.0469 E | 43.83      | —             | 12.8   |
| Trip 4 (2024)                   | 19                       | 29/08/2024 | 13:52 | 19.8123 N, 48.2405 E | 42.83      | —             | 14.1   |

Continued on next page

| Expedition    | Site | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------|------------|-------|----------------------|------------|---------------|--------|
| Trip 4 (2024) | 20   | 29/08/2024 | 13:57 | 19.8693 N, 48.4246 E | 41.94      | —             | 14.5   |
| Trip 4 (2024) | 21   | 29/08/2024 | 14:32 | 19.9011 N, 48.4865 E | 41.72      | —             | 13.1   |
| Trip 4 (2024) | 22   | 29/08/2024 | 14:50 | 19.9787 N, 48.6394 E | 43.33      | —             | 13     |
| Trip 4 (2024) | 23   | 29/08/2024 | 15:21 | 20.0493 N, 48.7733 E | 44.28      | —             | 12     |
| Trip 4 (2024) | 24   | 29/08/2024 | 15:55 | 20.0927 N, 48.8994 E | 43.17      | —             | 12.7   |
| Trip 4 (2024) | 25   | 29/08/2024 | 16:45 | 20.1003 N, 48.9542 E | 43.5       | —             | 11.6   |
| Trip 4 (2024) | 26   | 29/08/2024 | 17:02 | 20.1488 N, 49.0787 E | 42.83      | —             | 12.3   |
| Trip 4 (2024) | 27   | 29/08/2024 | 17:28 | 20.2039 N, 49.1960 E | 41.83      | —             | 12.6   |
| Trip 4 (2024) | 28   | 29/08/2024 | 17:56 | 20.2438 N, 49.3709 E | 40.83      | —             | 13.3   |
| Trip 4 (2024) | 29   | 29/08/2024 | 18:21 | 20.2598 N, 49.5675 E | 40.78      | —             | 13.3   |
| Trip 4 (2024) | 30   | 30/08/2024 | 05:51 | 20.2693 N, 49.6837 E | 36.5       | —             | 27.7   |
| Trip 4 (2024) | 31   | 30/08/2024 | 06:08 | 20.2946 N, 49.8914 E | 35.89      | —             | 27.3   |
| Trip 4 (2024) | 32   | 30/08/2024 | 06:30 | 20.3107 N, 50.0930 E | 35.72      | —             | 21.9   |
| Trip 4 (2024) | 33   | 30/08/2024 | 06:50 | 20.3175 N, 50.3031 E | 35.89      | —             | 23     |
| Trip 4 (2024) | 34   | 30/08/2024 | 07:11 | 20.3444 N, 50.5154 E | 36.39      | —             | 23.5   |
| Trip 4 (2024) | 35   | 30/08/2024 | 07:25 | 20.3718 N, 50.7300 E | 38.5       | —             | 21.2   |
| Trip 4 (2024) | 36   | 30/08/2024 | 08:01 | 20.4232 N, 51.1448 E | 39.28      | —             | 31.8   |
| Trip 4 (2024) | 37   | 30/08/2024 | 08:47 | 20.4542 N, 51.3449 E | 39.44      | —             | 23.2   |
| Trip 4 (2024) | 38   | 30/08/2024 | 09:14 | 20.4610 N, 51.3702 E | 41.06      | —             | 20     |
| Trip 4 (2024) | 39   | 30/08/2024 | 09:20 | 20.5102 N, 51.5974 E | 42.44      | —             | 19.5   |
| Trip 4 (2024) | 40   | 30/08/2024 | 09:49 | 20.5472 N, 51.8256 E | 42.06      | —             | 19     |
| Trip 4 (2024) | 41   | 30/08/2024 | 10:25 | 20.5398 N, 51.9461 E | 43.17      | —             | 17     |
| Trip 4 (2024) | 42   | 30/08/2024 | 10:50 | 20.5552 N, 52.1240 E | 43.67      | —             | 16.2   |
| Trip 4 (2024) | 43   | 30/08/2024 | 11:19 | 20.5676 N, 52.2241 E | 42.78      | —             | 17     |
| Trip 4 (2024) | 44   | 30/08/2024 | 11:40 | 20.5678 N, 52.4104 E | 41.94      | —             | 18     |
| Trip 4 (2024) | 45   | 30/08/2024 | 12:12 | 20.5635 N, 52.4678 E | 40.94      | —             | 17.1   |
| Trip 4 (2024) | 46   | 30/08/2024 | 12:22 | 20.5537 N, 52.6692 E | 44         | —             | 15.7   |
| Trip 4 (2024) | 47   | 30/08/2024 | 13:16 | 20.5418 N, 52.8688 E | 42.94      | —             | 15.3   |
| Trip 4 (2024) | 48   | 30/08/2024 | 13:57 | 20.5686 N, 53.0940 E | 42.5       | —             | 18.2   |
| Trip 4 (2024) | 49   | 30/08/2024 | 14:28 | 20.6447 N, 53.2879 E | 44         | —             | 14.4   |
| Trip 4 (2024) | 50   | 31/08/2024 | 06:08 | 20.7139 N, 53.4123 E | 35.61      | —             | 25.1   |
| Trip 4 (2024) | 51   | 31/08/2024 | 06:34 | 20.7987 N, 53.4870 E | 36.11      | —             | 25     |
| Trip 4 (2024) | 52   | 31/08/2024 | 06:54 | 20.8278 N, 53.5784 E | 36.28      | —             | 25.5   |
| Trip 4 (2024) | 53   | 31/08/2024 | 09:00 | 20.8515 N, 53.7579 E | 42.61      | —             | 17.5   |
| Trip 4 (2024) | 54   | 31/08/2024 | 09:34 | 20.9202 N, 53.9622 E | 42.5       | —             | 17.5   |

Continued on next page

| Expedition    | Site | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------|------------|-------|----------------------|------------|---------------|--------|
| Trip 4 (2024) | 55   | 31/08/2024 | 09:48 | 20.9023 N, 54.0024 E | 40.72      | —             | 20     |
| Trip 4 (2024) | 56   | 31/08/2024 | 10:16 | 20.8216 N, 54.2239 E | 40.44      | —             | 21.4   |
| Trip 4 (2024) | 57   | 31/08/2024 | 10:40 | 20.7350 N, 54.4176 E | 42.78      | —             | 18     |
| Trip 4 (2024) | 58   | 31/08/2024 | 10:58 | 20.6654 N, 54.5933 E | 43.5       | —             | 15     |
| Trip 4 (2024) | 59   | 31/08/2024 | 11:06 | 20.6563 N, 54.6359 E | 43         | —             | 14.9   |
| Trip 4 (2024) | 60   | 31/08/2024 | 11:28 | 20.7064 N, 54.7321 E | 42.89      | —             | 18.1   |
| Trip 5 (2025) | 1    | 12/10/2025 | 07:32 | 19.1498 N, 45.1665 E | 26.9       | 894           | 24.3   |
| Trip 5 (2025) | 2    | 12/10/2025 | 08:05 | 19.2035 N, 45.2448 E | 27.8       | 902           | 23.4   |
| Trip 5 (2025) | 3    | 12/10/2025 | 08:38 | 19.3011 N, 45.3844 E | 29         | 912           | 23     |
| Trip 5 (2025) | 4    | 12/10/2025 | 09:20 | 19.3796 N, 45.5518 E | 32.4       | 922           | 18.8   |
| Trip 5 (2025) | 5    | 12/10/2025 | 10:00 | 19.4072 N, 45.6595 E | 33.4       | 924           | 19.2   |
| Trip 5 (2025) | 6    | 12/10/2025 | 10:50 | 19.4057 N, 45.8161 E | 36.7       | 916           | 14.1   |
| Trip 5 (2025) | 7    | 12/10/2025 | 11:18 | 19.4070 N, 45.9539 E | 35         | 918           | 13.4   |
| Trip 5 (2025) | 8    | 12/10/2025 | 11:55 | 19.3932 N, 46.1107 E | 34         | 918           | 12.9   |
| Trip 5 (2025) | 9    | 12/10/2025 | 12:33 | 19.4112 N, 46.3027 E | 35.5       | 918           | 12.5   |
| Trip 5 (2025) | 10   | 12/10/2025 | 12:58 | 19.4205 N, 46.4608 E | 35.6       | 920           | 10.5   |
| Trip 5 (2025) | 11   | 12/10/2025 | 13:30 | 19.4444 N, 46.6356 E | 35.6       | 926           | 10.7   |
| Trip 5 (2025) | 12   | 12/10/2025 | 13:57 | 19.4779 N, 46.8344 E | 36.3       | 928           | 10.3   |
| Trip 5 (2025) | 13   | 12/10/2025 | 15:25 | 19.5293 N, 47.0533 E | 36.5       | 934           | 10.1   |
| Trip 5 (2025) | 14   | 12/10/2025 | 15:51 | 19.5813 N, 47.2920 E | 35         | 936           | 11.9   |
| Trip 5 (2025) | 15   | 12/10/2025 | 16:22 | 19.6137 N, 47.4759 E | 34.7       | 940           | 11.9   |
| Trip 5 (2025) | 16   | 12/10/2025 | 16:42 | 19.6445 N, 47.6261 E | 33.9       | 942           | 13.2   |
| Trip 5 (2025) | 17   | 13/10/2025 | 07:05 | 19.6765 N, 47.8273 E | 26.4       | 948           | 32.1   |
| Trip 5 (2025) | 18   | 13/10/2025 | 07:42 | 19.7325 N, 48.0469 E | 26.6       | 950           | 27     |
| Trip 5 (2025) | 19   | 13/10/2025 | 08:15 | 19.8123 N, 48.2405 E | 28.2       | 954           | 27.5   |
| Trip 5 (2025) | 20   | 13/10/2025 | 09:04 | 19.8693 N, 48.4246 E | 32.9       | 958           | 23.5   |
| Trip 5 (2025) | 21   | 13/10/2025 | 09:30 | 19.9011 N, 48.4865 E | 33         | 958           | 25     |
| Trip 5 (2025) | 22   | 13/10/2025 | 10:10 | 19.9787 N, 48.6394 E | 35.5       | 962           | 20.8   |
| Trip 5 (2025) | 23   | 13/10/2025 | 10:56 | 20.0493 N, 48.7733 E | 36.1       | 960           | 21.6   |
| Trip 5 (2025) | 24   | 13/10/2025 | 11:39 | 20.0927 N, 48.8994 E | 36.3       | 958           | 18.2   |
| Trip 5 (2025) | 25   | 13/10/2025 | 11:55 | 20.1003 N, 48.9542 E | 35         | 960           | 18.4   |
| Trip 5 (2025) | 26   | 13/10/2025 | 12:16 | 20.1488 N, 49.0787 E | 34.3       | 960           | 19.5   |
| Trip 5 (2025) | 27   | 13/10/2025 | 12:42 | 20.2039 N, 49.1960 E | 36.4       | 960           | 16.6   |
| Trip 5 (2025) | 28   | 13/10/2025 | 13:08 | 20.2438 N, 49.3709 E | 38.8       | 960           | 15.5   |
| Trip 5 (2025) | 29   | 13/10/2025 | 13:56 | 20.2598 N, 49.5675 E | 37.5       | 966           | 14.9   |

Continued on next page

| Expedition    | Site       | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------------|------------|-------|----------------------|------------|---------------|--------|
| Trip 5 (2025) | 30         | 13/10/2025 | 14:13 | 20.2693 N, 49.6837 E | 37.5       | 966           | 14.2   |
| Trip 5 (2025) | 31         | 13/10/2025 | 14:46 | 20.2946 N, 49.8914 E | 35.9       | 968           | 17.5   |
| Trip 5 (2025) | 32         | 13/10/2025 | 15:38 | 20.3107 N, 50.0930 E | 39         | 970           | 12.8   |
| Trip 5 (2025) | 33         | 13/10/2025 | 16:10 | 20.3175 N, 50.3031 E | 37.9       | 972           | 11.9   |
| Trip 5 (2025) | 34         | 13/10/2025 | 16:58 | 20.3444 N, 50.5154 E | 34.3       | 974           | 17.9   |
| Trip 5 (2025) | 35         | 14/10/2025 | 06:58 | 20.3718 N, 50.7300 E | 35.1       | 982           | 27.1   |
| Trip 5 (2025) | 36         | 14/10/2025 | 08:36 | 20.4232 N, 51.1448 E | 33.2       | 982           | 22.6   |
| Trip 5 (2025) | 37         | 14/10/2025 | 09:01 | 20.4542 N, 51.3449 E | 32.3       | 984           | 23.7   |
| Trip 5 (2025) | 38         | 14/10/2025 | 09:13 | 20.4610 N, 51.3702 E | 33.2       | 984           | 22.3   |
| Trip 5 (2025) | 39         | 14/10/2025 | 09:50 | 20.5102 N, 51.5974 E | 37.2       | 986           | 22.1   |
| Trip 5 (2025) | 40         | 14/10/2025 | 10:40 | 20.5472 N, 51.8256 E | 39         | 984           | —      |
| Trip 5 (2025) | 41         | 14/10/2025 | 11:04 | 20.5398 N, 51.9461 E | 37.1       | 986           | 20.8   |
| Trip 5 (2025) | 42         | 14/10/2025 | 11:40 | 20.5552 N, 52.1240 E | 36         | 984           | 22.6   |
| Trip 5 (2025) | 43         | 14/10/2025 | 12:07 | 20.5676 N, 52.2241 E | 40.4       | 984           | 19.7   |
| Trip 5 (2025) | 44         | 14/10/2025 | 13:03 | 20.5678 N, 52.4104 E | 37.1       | 984           | 19.3   |
| Trip 5 (2025) | 45         | 14/10/2025 | 13:16 | 20.5635 N, 52.4678 E | 39.5       | 984           | 17.6   |
| Trip 5 (2025) | 46         | 14/10/2025 | 14:02 | 20.5537 N, 52.6692 E | 38.6       | 982           | 17.2   |
| Trip 5 (2025) | 47         | 14/10/2025 | 14:41 | 20.5418 N, 52.8688 E | 36.2       | 984           | 23     |
| Trip 5 (2025) | 48         | 14/10/2025 | 15:30 | 20.5686 N, 53.0940 E | 36.7       | 986           | 19.2   |
| Trip 5 (2025) | 49         | 14/10/2025 | 16:20 | 20.6447 N, 53.2879 E | 35.2       | 986           | 20.4   |
| Trip 5 (2025) | 50         | 14/10/2025 | 17:00 | 20.7139 N, 53.4123 E | 34.7       | 984           | 20.9   |
| Trip 5 (2025) | 51         | 15/10/2025 | 06:41 | 20.7987 N, 53.4870 E | 31.1       | 990           | 36.3   |
| Trip 5 (2025) | 52         | 15/10/2025 | 07:09 | 20.8278 N, 53.5784 E | 33.5       | 990           | 34.1   |
| Trip 5 (2025) | 53         | 15/10/2025 | 07:59 | 20.8515 N, 53.7579 E | 33.7       | 990           | 32.4   |
| Trip 5 (2025) | 54         | 15/10/2025 | 09:14 | 20.9202 N, 53.9622 E | 37.5       | 992           | 24.2   |
| Trip 5 (2025) | 55         | 15/10/2025 | 09:59 | 20.9023 N, 54.0024 E | 39.3       | 988           | 21.4   |
| Trip 5 (2025) | 56         | 15/10/2025 | 10:38 | 20.8216 N, 54.2239 E | 42.8       | 988           | 19.8   |
| Trip 5 (2025) | 57         | 15/10/2025 | 11:18 | 20.7350 N, 54.4176 E | 39.4       | 986           | 20.3   |
| Trip 5 (2025) | 58         | 15/10/2025 | 11:41 | 20.6654 N, 54.5933 E | 41.8       | 986           | 18.3   |
| Trip 5 (2025) | 59         | 15/10/2025 | 11:59 | 20.6563 N, 54.6359 E | 40.2       | 984           | 20.7   |
| Trip 5 (2025) | 60         | 15/10/2025 | 13:55 | 20.7064 N, 54.7321 E | 35.9       | 988           | 25.5   |
| Trip 5 (2025) | oill spill | 12/10/2025 | 10:20 | 19.4054 N, 45.7271 E | 34.1       | 924           | 17.1   |
| Trip 5 (2025) | hot spring | 14/10/2025 | 07:58 | 20.3972 N, 50.9655 E | 32.7       | 984           | 22.8   |

*Metadata QC.* For Trip 2, the eight values 34.5–41.9 occur in the source humidity column while the temperature column is empty. The ledger assigns them to temperature; neither pressure nor relative humidity is inferred. Trip 3 site 21 is retained as 31.321% RH because

the original workbook records the fraction 0.31321. The 15 March records appended to the legacy Trip 3 worksheet are dated 2023 in that workbook; the curated table therefore labels them Trip 1 auxiliary/revisit records, and Trip 3 ends on 21 February 2024. [The Trip 1 and Trip 3 Site 52 rows \(and the appended 52AD record\) repeat the Site 53 coordinates; the ledger restores the Site 52 location recorded in Trips 4 and 5 \(20.8278°N, 53.5784°E\), which the Trip 3 GPS track corroborates.](#) Trip 5 site 40 contains the source value 194% RH, which is outside the physical range. Because no primary source currently establishes a decimal correction, the curated value is missing; 194 is preserved in the correction ledger.

**Table S3:** Audited XRF reporting-channel cross-references. Labels, formulas, charge and entity types were checked against pinned ChEBI and PubChem records.

| Analyte | Formula/type | ChEBI       | PubChem     |
|---------|--------------|-------------|-------------|
| Ag      | Ag           | CHEBI:30512 | CID:23954   |
| Al      | Al           | CHEBI:28984 | CID:5359268 |
| Al2O3   | Al2O3        | CHEBI:30187 | CID:9989226 |
| As      | As           | CHEBI:27563 | CID:5359596 |
| Au      | Au           | CHEBI:29287 | CID:23985   |
| Ba      | Ba           | CHEBI:32594 | CID:5355457 |
| BaO     | BaO          | –           | CID:62392   |
| Bi      | Bi           | CHEBI:33301 | CID:5359367 |
| Br      | Br           | CHEBI:22927 | CID:5360770 |
| Ca      | Ca           | CHEBI:22984 | CID:5460341 |
| CaO     | CaO          | CHEBI:31344 | CID:14778   |
| Cd      | Cd           | CHEBI:22977 | CID:23973   |
| Ce      | Ce           | CHEBI:33369 | CID:23974   |
| CeO2    | CeO2         | CHEBI:79089 | CID:73963   |
| Cl      | Cl           | CHEBI:23116 | CID:5360523 |
| Co      | Co           | CHEBI:27638 | CID:104730  |
| CoO     | CoO          | –           | CID:14786   |
| Cr      | Cr           | CHEBI:28073 | CID:23976   |
| Cr2O3   | Cr2O3        | CHEBI:48242 | CID:517277  |
| Cs      | Cs           | CHEBI:30514 | CID:5354618 |
| Cu      | Cu           | CHEBI:28694 | CID:23978   |
| CuO     | CuO          | CHEBI:75955 | CID:164827  |
| Dy      | Dy           | CHEBI:33377 | CID:23912   |
| Er      | Er           | CHEBI:33379 | CID:23980   |
| Eu      | Eu           | CHEBI:32999 | CID:23981   |
| Fe      | Fe           | CHEBI:18248 | CID:23925   |

| Analyte | Formula/type             | ChEBI       | PubChem     |
|---------|--------------------------|-------------|-------------|
| Fe2O3   | Fe2O3                    | CHEBI:50819 | CID:518696  |
| Ga      | Ga                       | CHEBI:49631 | CID:5360835 |
| Ga2O3   | Ga2O3                    | –           | CID:158605  |
| Gd      | Gd                       | CHEBI:33375 | CID:23982   |
| Ge      | Ge                       | CHEBI:30441 | CID:6326954 |
| Hf      | Hf                       | CHEBI:33343 | CID:23986   |
| Hg      | Hg                       | CHEBI:25195 | CID:23931   |
| Ho      | Ho                       | CHEBI:49648 | CID:23988   |
| I       | I                        | CHEBI:24859 | CID:5360629 |
| Ir      | Ir                       | CHEBI:49666 | CID:23924   |
| K       | K                        | CHEBI:26216 | CID:5462222 |
| K2O     | K2O                      | CHEBI:88321 | CID:9989219 |
| LE      | instrument_pseudoanalyte | –           | –           |
| La      | La                       | CHEBI:33336 | CID:23926   |
| La2O3   | La2O3                    | –           | CID:150906  |
| Lu      | Lu                       | CHEBI:33382 | CID:23929   |
| Mg      | Mg                       | CHEBI:25107 | CID:5462224 |
| MgO     | MgO                      | CHEBI:31794 | CID:14792   |
| Mn      | Mn                       | CHEBI:18291 | CID:23930   |
| MnO     | MnO                      | –           | CID:14940   |
| Mo      | Mo                       | CHEBI:28685 | CID:23932   |
| Na      | Na                       | CHEBI:26708 | CID:5360545 |
| Na2O    | Na2O                     | –           | CID:73971   |
| Nb      | Nb                       | CHEBI:33344 | CID:23936   |
| Nd      | Nd                       | CHEBI:33372 | CID:23934   |
| Nd2O3   | Nd2O3                    | –           | CID:159373  |
| Ni      | Ni                       | CHEBI:28112 | CID:935     |
| NiO     | NiO                      | –           | CID:14805   |

| Analyte | Formula/type | ChEBI       | PubChem     |
|---------|--------------|-------------|-------------|
| Os      | Os           | CHEBI:30687 | CID:23937   |
| P       | P            | CHEBI:28659 | CID:5462309 |
| P2O5    | P2O5         | CHEBI:37376 | CID:6326975 |
| Pb      | Pb           | CHEBI:25016 | CID:5352425 |
| PbO     | PbO          | CHEBI:81045 | CID:14827   |
| Pr      | Pr           | CHEBI:49828 | CID:23942   |
| PrO2    | PrO2         | –           | –           |
| Pt      | Pt           | CHEBI:33364 | CID:23939   |
| Rb      | Rb           | CHEBI:33322 | CID:5357696 |
| Re      | Re           | CHEBI:49882 | CID:23947   |
| S       | S            | CHEBI:26833 | CID:5362487 |
| SO3     | SO3          | CHEBI:29384 | CID:24682   |
| Sb      | Sb           | CHEBI:30513 | CID:5354495 |
| Sc      | Sc           | CHEBI:33330 | CID:23952   |
| Sc2O3   | Sc2O3        | –           | CID:134661  |
| Se      | Se           | CHEBI:27568 | CID:6326970 |
| Si      | Si           | CHEBI:27573 | CID:5461123 |
| SiO2    | SiO2         | CHEBI:30563 | CID:24261   |
| Sm      | Sm           | CHEBI:33374 | CID:23951   |
| Sm2O3   | Sm2O3        | –           | CID:159425  |
| Sn      | Sn           | CHEBI:27007 | CID:5352426 |
| Sr      | Sr           | CHEBI:33324 | CID:5359327 |
| SrO     | SrO          | –           | CID:73975   |
| Ta      | Ta           | CHEBI:33348 | CID:23956   |
| Tb      | Tb           | CHEBI:33376 | CID:23958   |
| Th      | Th           | CHEBI:33385 | CID:23960   |
| Ti      | Ti           | CHEBI:33341 | CID:23963   |
| TiO2    | TiO2         | CHEBI:32234 | CID:26042   |

| Analyte | Formula/type | ChEBI       | PubChem     |
|---------|--------------|-------------|-------------|
| Tm      | Tm           | CHEBI:33380 | CID:23961   |
| U       | U            | CHEBI:27214 | CID:23989   |
| V       | V            | CHEBI:27698 | CID:23990   |
| V2O5    | V2O5         | CHEBI:30045 | CID:14814   |
| W       | W            | CHEBI:27998 | CID:23964   |
| Y       | Y            | CHEBI:33331 | CID:23993   |
| Yb      | Yb           | CHEBI:33381 | CID:23992   |
| Zn      | Zn           | CHEBI:27363 | CID:23994   |
| ZnO     | ZnO          | CHEBI:36560 | CID:3007857 |
| Zr      | Zr           | CHEBI:33342 | CID:23995   |
| ZrO2    | ZrO2         | —           | CID:62395   |

*Interpretation.* A dash denotes an audited null, ~~not a missing review~~. These `rdfs:seeAlso` cross-references identify the element or oxide-equivalent reporting channel; ~~they do not assert that the corresponding molecule or mineral occurs in a sample~~. The P<sub>2</sub>O<sub>5</sub> channel is cross-referenced to the ChEBI P<sub>4</sub>O<sub>10</sub> class only because the formulas share the P:O empirical ratio. LE is an instrument aggregate, and PrO<sub>2</sub> has no formula-matching external identifier in the audited sources.

## References

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- [2] Degtyarenko, K., Matos, P., Ennis, M., Hastings, J., Zbinden, M., McNaught, A., Alcántara, R., Darsow, M., Guedj, M., Ashburner, M.: ChEBI: a database and ontology for chemical entities of biological interest. *Nucleic Acids Research* **36**(suppl.1), 344–350 (2008) <https://doi.org/10.1093/nar/gkm791>
- [3] Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B.A., Thiessen, P.A., Yu, B., *et al.*: PubChem 2023 update. *Nucleic Acids Research* **51**(D1), 1373–1380 (2023)